Integrating with inverse trig – 0.5.7 / 1.7

$$\frac{d}{dx}\arcsin(x) = \frac{1}{\sqrt{1-x^2}} \implies \int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin(x) + C$$

$$\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2} \implies \int \frac{1}{1+x^2}\,dx = \arctan(x) + C$$

Example: Find $\int \frac{1}{4+x^2}\,dx$

$$= \frac{1}{4}\int \frac{1}{1+\frac{x^2}{4}}\,dx = \frac{1}{4}\int \frac{1}{1+\left(\frac{x}{2}\right)^2}\,dx$$

$u = \frac{x}{2}$

$du = \frac{1}{2}\,dx \qquad dx = 2du$

$$= \frac{2}{4}\int \frac{1}{1+u^2}\,du = \frac{1}{2}\arctan(u) + C$$

$$= \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$$

$$\boxed{\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C}$$

Example :

$$\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \int \frac{1}{\sqrt{a^2\left(1 - \frac{x^2}{a^2}\right)}}\,dx$$

$$= \frac{1}{a} \int \frac{1}{\sqrt{1 - \left(\frac{x}{a}\right)^2}}\,dx$$

$u = \frac{x}{a}$

$du = \frac{1}{a}dx$

$dx = a\,du$

$$= \frac{a}{a} \int \frac{1}{\sqrt{1 - u^2}}\,du$$

$$= \arcsin(u) + C = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + C$$

Example: $\int \frac{x}{\sqrt{1-x^4}}\,dx$ $\quad u = x^2$, $du = 2x\,dx$

$$= \frac{1}{2}\int \frac{1}{\sqrt{1-u^2}}\,du = \frac{1}{2}\arcsin(u) + C = \frac{1}{2}\arcsin(x^2) + C$$

Example: $\int \frac{1}{x^2+2x+5}\,dx$

Complete the square: x^2+2x+5

$$= \underbrace{x^2+2x+1}_{=(x+1)^2} - 1 + 5 = (x+1)^2 + 4$$

$$= \int \frac{1}{4+(x+1)^2}\,dx \qquad u = x+1, \quad du = dx \qquad = \int \frac{1}{4+u^2}\,du = \frac{1}{2}\arctan\left(\frac{u}{2}\right) + C$$

$$= \frac{1}{2}\arctan\left(\frac{x+1}{2}\right) + C$$